Name: _____

TOPIC TEST 1

Chapter 11: Exponential functions

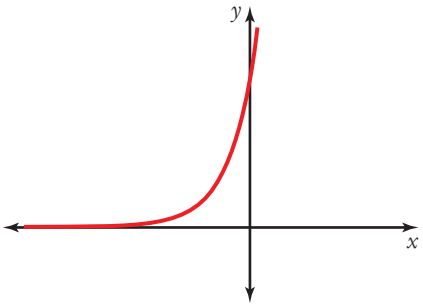
- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 13 short answer questions (27 marks)
- Part C: 7 analysis questions (22 marks)

Part A

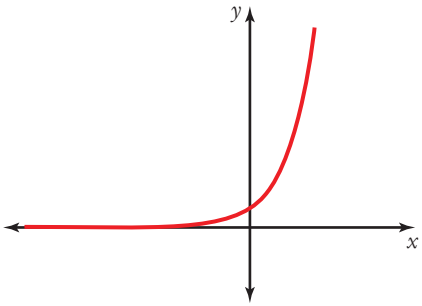
5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

Use the graphs below to answer questions **1** and **2**.

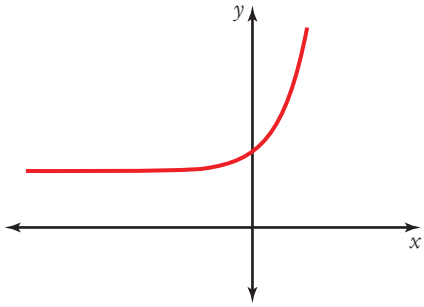
A



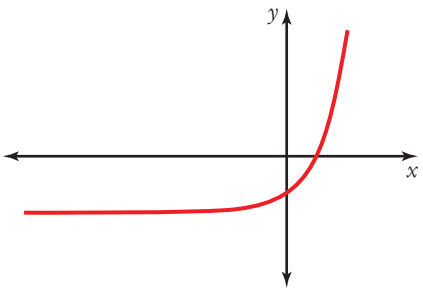
B



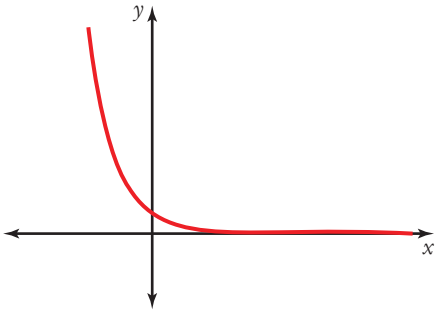
C



D



E



© Cengage Learning Australia Pty Ltd 2013

1 Which of the graphs **A**–**E** above is most likely to represent $y = a^x + b$, where $a > 1$ and $b > 0$?

[1 mark]

2 Which of the graphs **A**–**E** above is most likely to represent $y = a^x + b$, where $0 < a < 1$ and $b = 0$?

[1 mark]

3 The solution of $2^{3x+2} - 256 = 0$ is:

A $x = 8$

B $x = 6$

C $x = 2$

D $x = -2$

E $x = -6$

[1 mark]

4 The table on the right shows some points of an exponential function $y = f(x)$.

x	1	0	2	3
$f(x)$	72	60	86.4	103.68

The initial value a and growth factor b are:

A $a = 72, b = 60$

B $a = 72, b = 1.2$

C $a = 1.2, b = 60$

D $a = 1.2, b = 72$

E $a = 60, b = 1.2$

[1 mark]

5 A radioactive sample decays by 5% every year. It has an initial activity of 85.7 mC (millicuries). After 4 years, the activity is about:

A 69.8 mC

B 68.6 mC

C 65.7 mC

D 104.2 mC

E 102.8 mC

[1 mark]

© Cengage Learning Australia Pty Ltd 2013

Part B

7 short answer questions
27 marks
Show your working where appropriate.

1 Solve $2x^2 - 3x - 1 = 0$ correct to 3 significant figures.

[2 marks]

2 Simplify and express with positive indices.

a $\left(4v^{-\frac{1}{2}}\right)^{1\frac{1}{2}} \times 5\sqrt{v}$

b $\dfrac{(2d^6k^3)^{-\frac{1}{3}} \times 8d^{\frac{3}{4}}k^{\frac{2}{3}}}{3d^{-\frac{1}{4}} \times \sqrt[3]{4k^4}}$

[6 marks]

3 $v(t) = 1.9 \times (0.7)^t + 2$. Find the value of:

a $v(5)$

b $v(-3)$

[2 marks]

4 Solve the following equations.

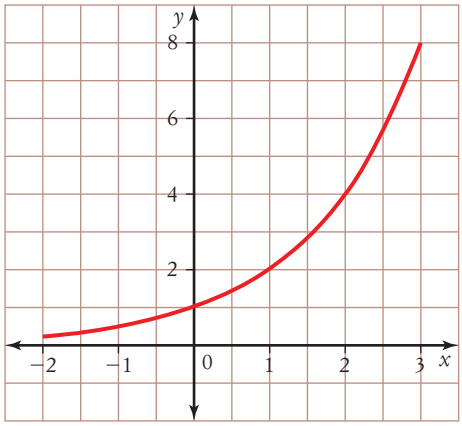
a $2^{x+1} = 32$

b $4^{3x+1} = 32$

c $64^{3x+1} = 32^{5x-4}$

[9 marks]

5 The graph of $y = 2^x$ is shown. Use the graph to solve $2^x = 6$



[2 marks]

6 Use a graph drawn with a CAS calculator to solve $16^x = 5$.

[2 marks]

7 For each of the following exponential functions, calculate the y-intercept and the equation of the horizontal asymptote.

a $y = 4 \times 2^{(x+3)}$

b $y = 2 \times 0.6^{(x+1)} - 5$

[4 marks]

© Cengage Learning Australia Pty Ltd 2013

Part C

4 analysis questions
22 marks
Show your working where appropriate.

1 Sketch the graph of each of the following by determining the key features of the graph.

a $y = 2^x - 5$

b $y = 3^{x-2} + 2$

[6 marks]

2 Sketch the graph of $y = 3^{x+3} - 5$ by transforming the graph of $y = 3^x$.

[5 marks]

3 The radioactivity of a sample is found to be 150 mCu (millicuries). The radioactive source decays by 2% a year, so the radioactivity decreases at this rate.

a Set up an exponential model of the radioactivity.

b Find the radioactivity of the sample after:

i 5 years _____

ii 10 years _____

iii 100 years _____

© Cengage Learning Australia Pty Ltd 2013

c Determine when the radioactivity will have decreased to half its initial value (the half-life).

[7 marks]

4 The population of rabbits in an area increases by an average of 50% every month. How long would it take two rabbits to increase to a population of 3000 rabbits and overrun an area?

[4 marks]
Total marks: 54

© Cengage Learning Australia Pty Ltd 2013

Answers

Part A [1 mark each]

1 The graph of $y = a^x + b$ will look like the graph of $y = a^x$ (where $a > 1$) that has been translated b units up because $b > 0$.

$\therefore$ **C** **[1 mark]**

2 The graph of $y = a^x + b$ will look like the graph of $y = a^x$ (where $0 < a < 1$) because $b = 0$.

$\therefore$ **E** **[1 mark]**

3 Since $256 = 2^8$, rewrite the equation as $2^{3x+2} - 2^8 = 0$ or $2^{3x+2} = 2^8$

Equating the indices, we get $3x + 2 = 8$. Solving for x, we obtain $x = 2$. Alternatively, put each of the options into the equation and see which one gives the correct answer.

$\therefore$ **C** **[1 mark]**

4 When $x = 0$, $f(x) = 60$, so the initial value $(a) = 60$.

$b = \dfrac{103.68}{86.4} = \dfrac{86.4}{72} = \dfrac{72}{60} = 1.2$

$\therefore$ **E** **[1 mark]**

5 $A = 85.7 \times (0.95)^t = 85.7 \times (0.95)^4 \approx 69.8$.

$\therefore$ **A** **[1 mark]**

Part B

1 $2x^2 - 3x - 1 = 0$ so $a = 2$, $b = -3$ and $c = -1$ **[1 mark]**

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{3 \pm \sqrt{9+8}}{4} = \frac{3 \pm \sqrt{17}}{4} = 1.7807\ldots \text{ or } -0.28077\ldots$$

$x \approx 1.78$ or -0.281 **[1 mark]**

[2 marks]

2 a $\left(4v^{-\frac{1}{2}}\right)^{1\frac{1}{2}} \times 5\sqrt{v} = 4^{\frac{3}{2}} v^{-\frac{3}{4}} \times 5v^{\frac{1}{2}} = 8v^{-\frac{1}{4}} \times 5 = \dfrac{40}{v^{\frac{1}{4}}}$ **[2 marks]**

b $$\frac{(2d^6k^3)^{-\frac{1}{3}} \times 8d^{\frac{3}{4}}k^{\frac{2}{3}}}{3d^{-\frac{1}{4}} \times \sqrt[3]{4k^4}} = \frac{2^{-\frac{1}{3}}d^{-2}k^{-1} \times 2^3 d^{\frac{3}{4}}k^{\frac{2}{3}}}{3d^{-\frac{1}{4}} \times 2^{\frac{2}{3}}k^{\frac{4}{3}}}$$

$$= \frac{2^{-\frac{1}{3}-\frac{2}{3}+3} \times d^{-2+\frac{3}{4}+\frac{1}{4}} k^{-1+\frac{2}{3}-\frac{4}{3}}}{3}$$

$$= \frac{2^2 d^{-1} k^{-\frac{5}{3}}}{3} = \frac{4}{3dk^{\frac{5}{3}}}$$

[3 marks]

[6 marks]

3 a $v(5) = 1.9 \times (0.7)^5 + 2 \approx 2.3193$ **[1 mark]**

b $v(-3) = 1.9 \times (0.7)^{-3} + 2 \approx 7.5394$ **[1 mark]**

[2 marks]

4 a $2^{x+1} = 32 \Rightarrow 2^{x+1} = 2^5$ **[1 mark]**

$\Rightarrow x + 1 = 5 \Rightarrow x = 4$ **[2 marks]**

b $4^{3x+1} = 32 \Rightarrow 2^{2(3x+1)} = 2^5$ **[1 mark]**

$\Rightarrow 6x + 2 = 5 \Rightarrow x = \frac{1}{2}$ **[2 marks]**

c $64^{3x+1} = 32^{5x-4} \Rightarrow 2^{6(3x+1)} = 2^{5(5x-4)}$ **[1 mark]**

$\Rightarrow 18x + 8 = 25x - 20 \Rightarrow x = 3\frac{5}{7}$ **[2 marks]**

[9 marks]

5 Locate the point on the graph where $y = 6$. **[1 mark]**

Draw a vertical line to intersect the x axis and read the value of x.

$x \approx 2.6$

[2 marks]

6 Draw the graph. Use the trace function or locate the zero of $16^x - 5 = 0$.

$x \approx 0.580$ **[2 marks]**

7 a Let $x = 0$: $y = 4 \times 2^{(0+3)} = 32$; y-intercept $= 32$ **[1 mark]**

Compare with $y = a^{x+B} + C$: $C = 0$, so horizontal asymptote is $y = 0$ **[1 mark]**

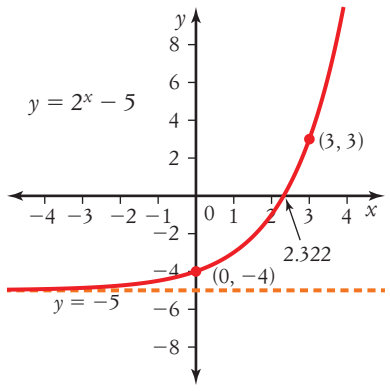
b Let $x = 0$: $y = 2 \times 0.6^{(0+1)} - 5 = -3.8$; y-intercept $= -3.8$ **[1 mark]**

Compare with $y = a^{x+B} + C$: $C = -5$, so the horizontal asymptote is $y = -5$ **[1 mark]**

[4 marks]

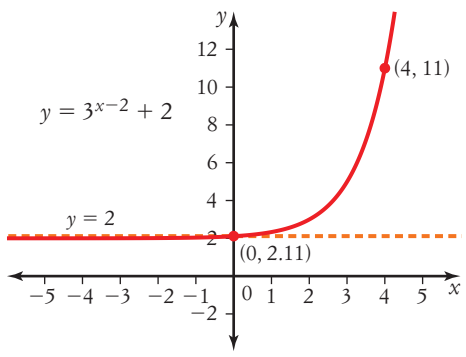
Part C

1 a



[3 marks]

b

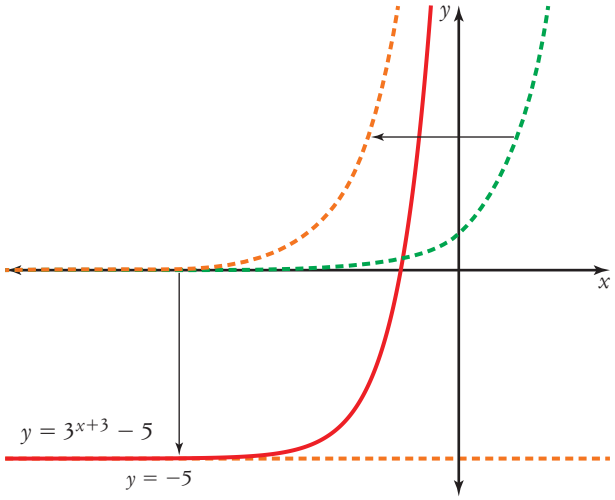


[3 marks]

[6 marks]

2 Compare $y = 3^{x+3} - 5$ with $y = a^{x+B} - C$. $B = 3$ and $C = -5$.

Plot the graph of $y = 3x$. Slide the graph 3 units to the left and then slide the graph 5 units down. **[2 marks]**



[3 marks]

[5 marks]

3 a Growth factor $= (100 - 2)\% = 0.98$; initial value $= 150$ **[1 mark]**

$R = 150 \times 0.98^{y}$ **[1 mark]**

b i $R = 150 \times 0.98^{5} \approx 135.5881$ mCu **[1 mark]**

ii $R = 150 \times 0.98^{10} \approx 122.5609$ mCu **[1 mark]**

iii $R = 150 \times 0.98^{100} \approx 19.8929$ mCu **[1 mark]**

c At half-life, $R = 75$ mCu

$75 = 150 \times 0.98^{y}$

$0.98^{y} = 0.5$ **[1 mark]**

Use trial and error to solve for y, or use the **solve()** function on your CAS calculator.

$y \approx 34.3$ years **[1 mark]**

[7 marks]

4 Growth factor $= (100 + 50)\% = 1.5$; initial value $= 2$ **[1 mark]**

Let m = number of months

Let P = population of rabbits

$P = 2 \times 1.5^{m} = 3000$ **[1 mark]**

$1500 = 1.5^{m}$ **[1 mark]**

Use trial and error to solve for m, or use the **solve()** function on your CAS calculator.

$m \approx 18.04$ months **[1 mark]**

[4 marks]